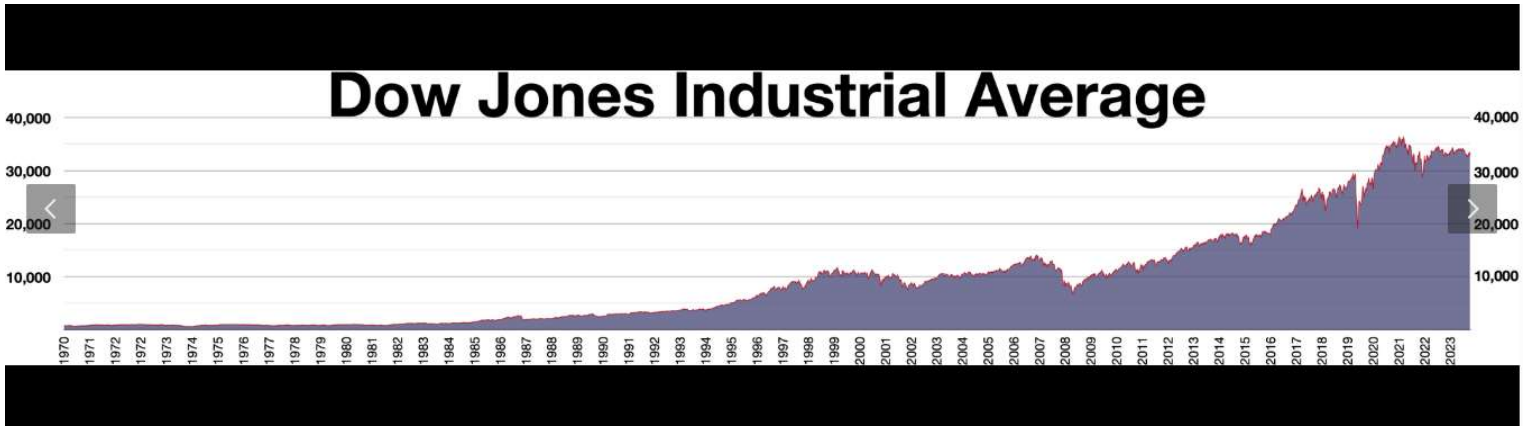
-DJI-
Dow Jones Industrial Average



ESİN FİRUZAN

23 Ekim 2026

Finansal Zaman Serisi Analizi

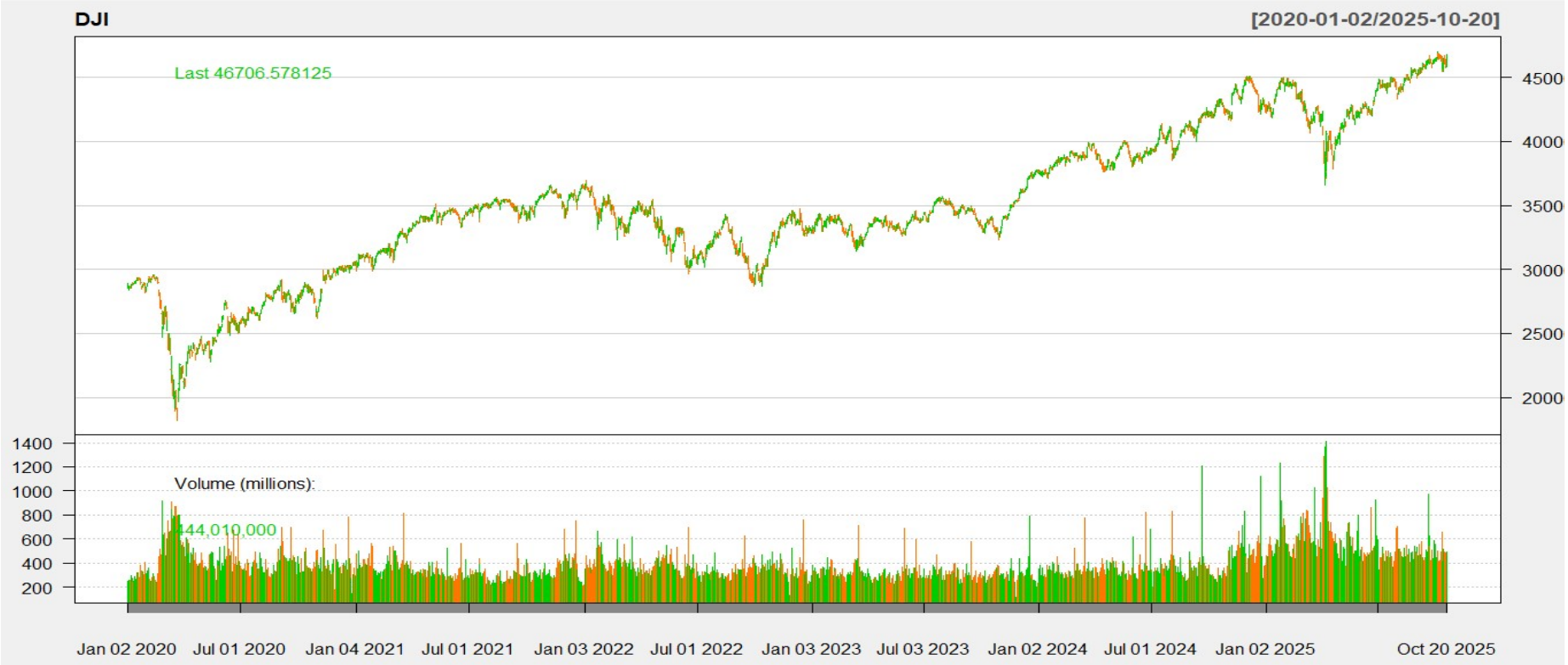
Ticari borsayı, menkul değer sembolleri ve endüstriyi gösteren DJIA bileşen şirketleri

Şirket	Hisse senedi	Sembol	Endüstri	Eklenen tarih	Notlar	Endeks ağırlıklandırma
3M	NYSE	NYSE: MMM	Konglomerat	1976-08-09	Minnesota Madencilik ve İmalat olarak	1,54%
American Express	NYSE	NYSE: AXP	Finansal hizmetler	1982-08-30		3,64%
Amgen	NASDAQ	AMGN	Biyofarmasötik	2020-08-31		4.80%
Amazon	NASDAQ	AMZN	Perakendecilik	2024-02-26		2.93%
Apple	NASDAQ	AAPL	Bilgi teknolojisi	2015-03-19		3,04%
Boeing	NYSE	NYSE: BA	Havacılık ve savunma	1987-03-12		3,36%
Caterpillar	NYSE	NYSE: CAT	İnşaat ve madencilik	1991-05-06		5,45%
Chevron	NYSE	NYSE: CVX	Petrol endüstrisi	2008-02-19	Ayrıca 1930-07-18 ile 1999-11-01 arası	2.59%
Cisco	NASDAQ	CSCO	Bilgi teknolojisi	2009-06-08		0,81%
Coca-Cola	NYSE	NYSE: KO	İçecek endüstrisi	1987-03-12	Ayrıca 1932-05-26 ila 1935-11-20	1,02%
Disney	NYSE	NYSE: DIS	Yayın ve eğlence	1991-05-06		1,81%
Dow	NYSE	NYSE: DOW	Kimya endüstrisi	1991-05-06		0,94%
Goldman Sachs	NYSE	NYSE: GS	Finansal hizmetler	2019-04-02		6,54%
Home Depot	NYSE	NYSE: HD	Ev tadilatı	1999-11-01		6,23%

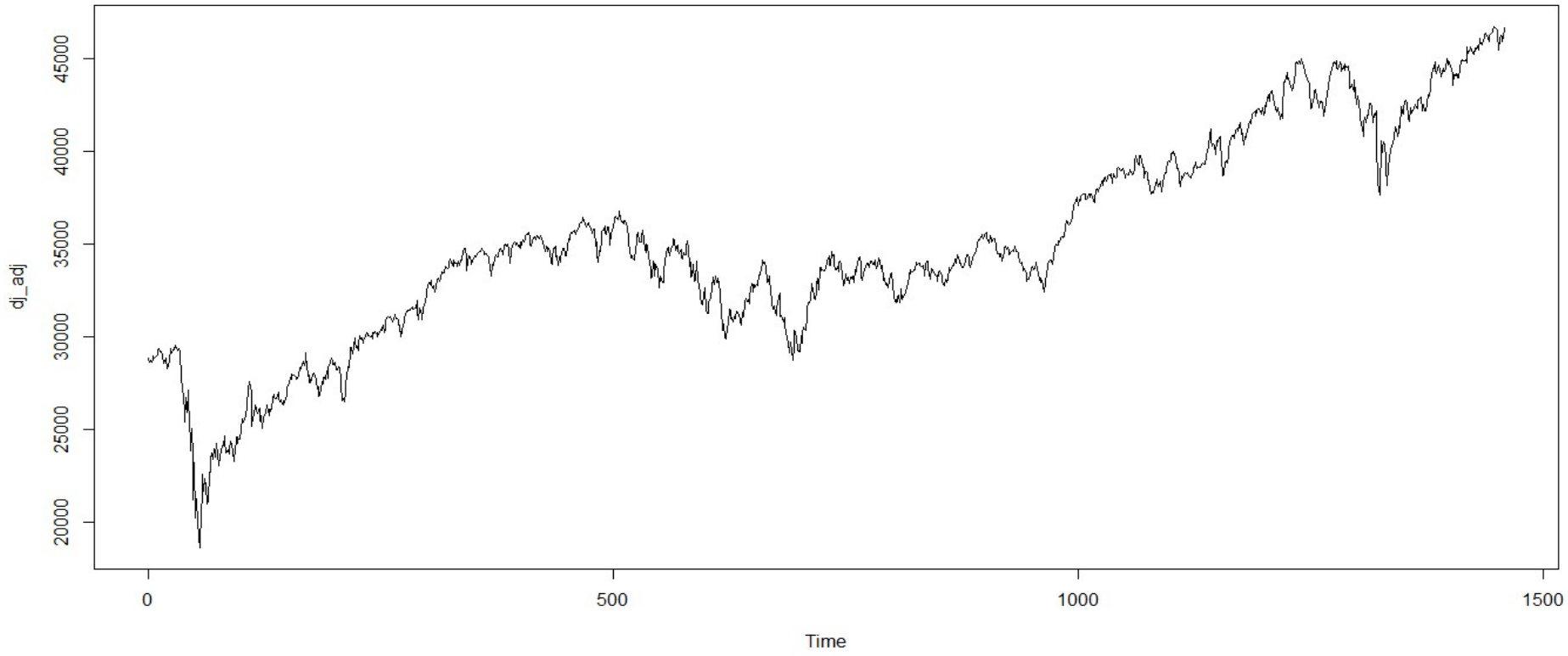
DJI

```
> head(DJI)
           DJI.Open DJI.High  DJI.Low DJI.Close DJI.Volume DJI.Adjusted
2020-01-02 28638.97 28872.80 28627.77  28868.80  251820000     28868.80
2020-01-03 28553.33 28716.31 28500.36  28634.88  239590000     28634.88
2020-01-06 28465.50 28708.02 28418.63  28703.38  252760000     28703.38
2020-01-07 28639.18 28685.50 28565.28  28583.68  258900000     28583.68
2020-01-08 28556.14 28866.18 28522.51  28745.09  291750000     28745.09
2020-01-09 28851.97 28988.01 28844.31  28956.90  275060000     28956.90
> tail(DJI)
           DJI.Open DJI.High  DJI.Low DJI.Close DJI.Volume DJI.Adjusted
2025-09-23 46364.11 46714.27 46217.58  46292.78  541530000     46292.78
2025-09-24 46368.94 46453.07 46096.39  46121.28  421040000     46121.28
2025-09-25 46097.43 46122.42 45785.17  45947.32  509180000     45947.32
2025-09-26 46101.45 46353.03 46051.92  46247.29  434270000     46247.29
2025-09-29 46306.34 46387.06 46149.05  46316.07  479740000     46316.07
2025-09-30 46282.63 46425.30 46103.39  46397.89  584780000     46397.89
> chartSeries(DJI, type = "bars", theme="white")
```

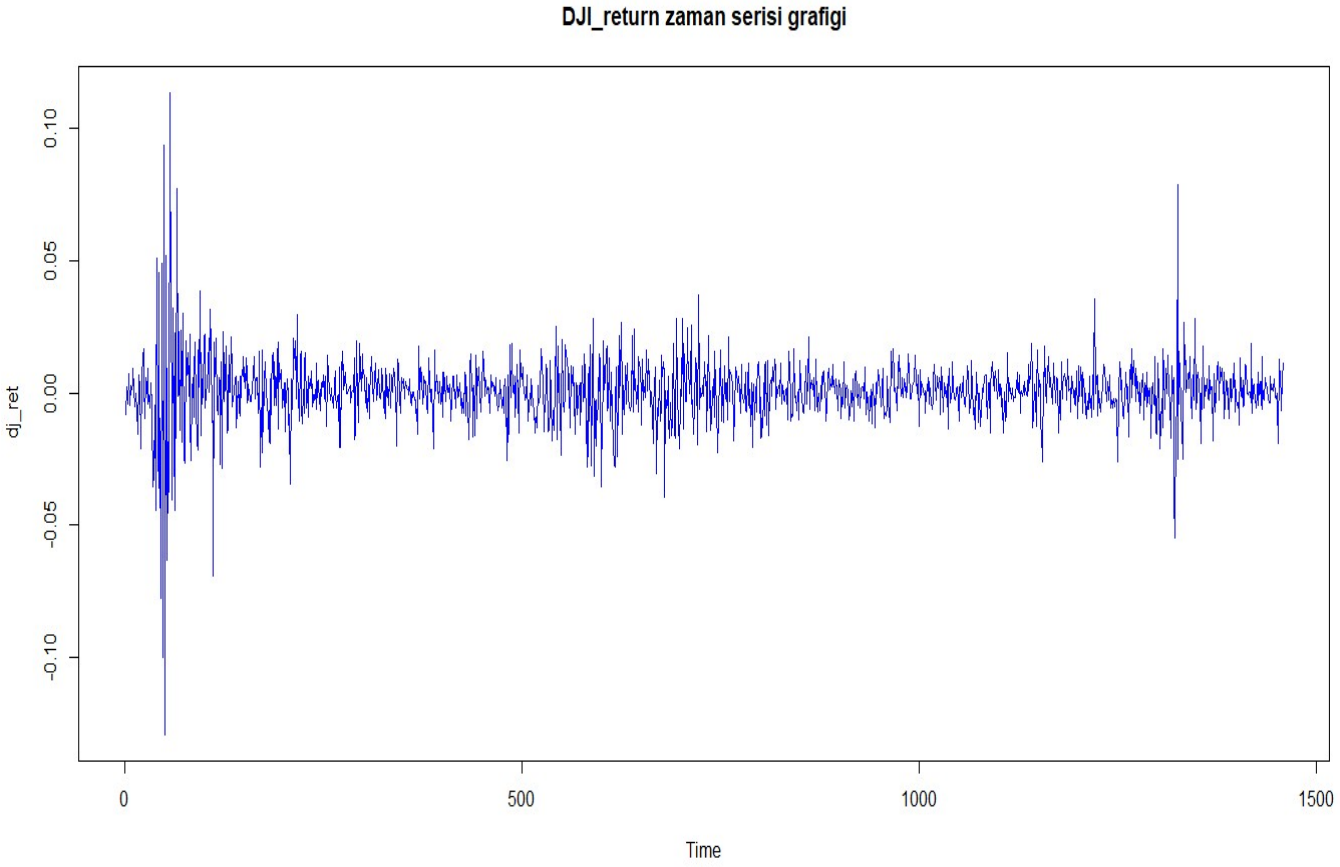
DJI_Adjusted



DJI-Adjusted



DJI_getiri

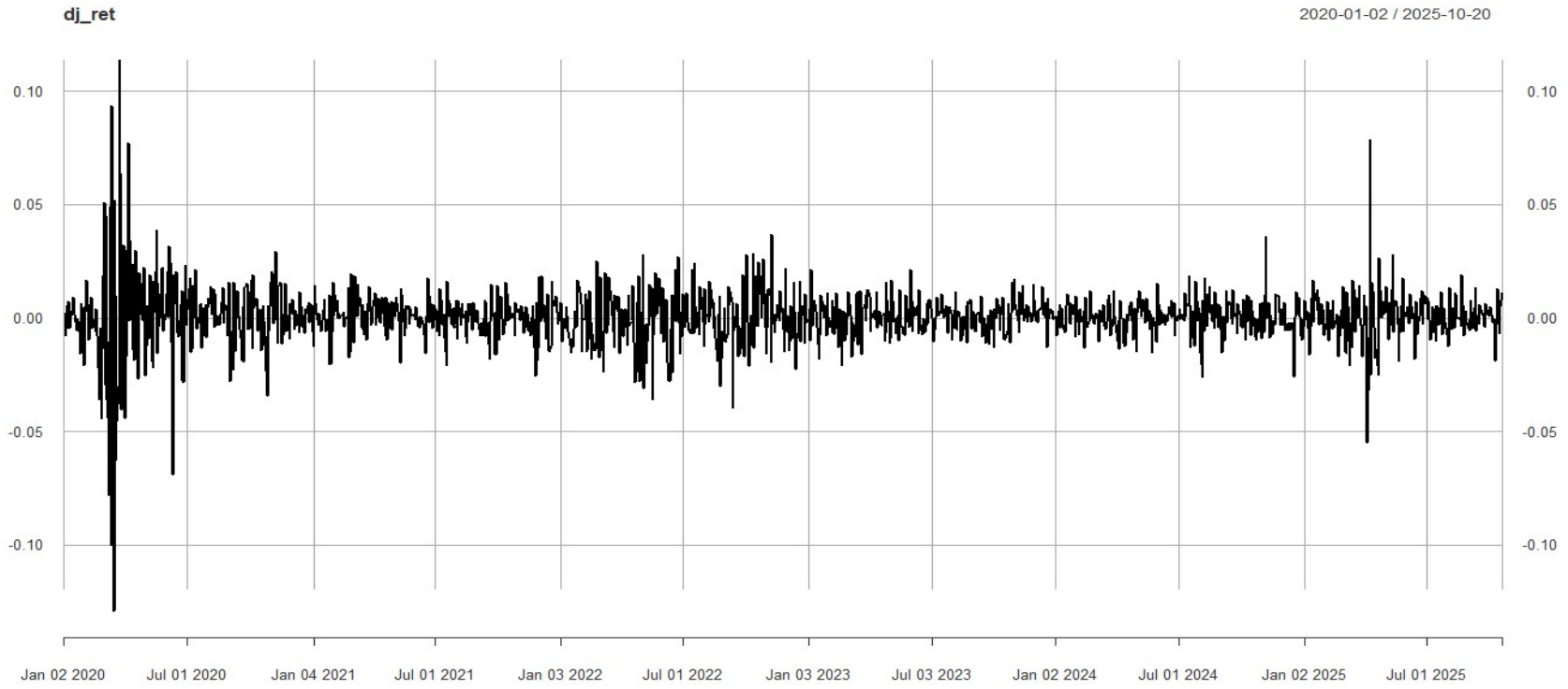


- Net Getiri (Simple Net return)

$$R_t = \frac{P_t}{P_{t-1}} - 1 = \frac{P_t - P_{t-1}}{P_{t-1}}$$

```
> head(dj_adj)
           DJI.Adjusted
2020-01-02     28868.80
2020-01-03     28634.88
2020-01-06     28703.38
2020-01-07     28583.68
2020-01-08     28745.09
2020-01-09     28956.90
> head(dj_ret)
           DJI.Adjusted
2020-01-03 -0.008102862
2020-01-06  0.002392187
2020-01-07 -0.004170281
2020-01-08  0.005646934
2020-01-09  0.007368582
2020-01-10 -0.004597552
```

Getiri DJI_adjusted

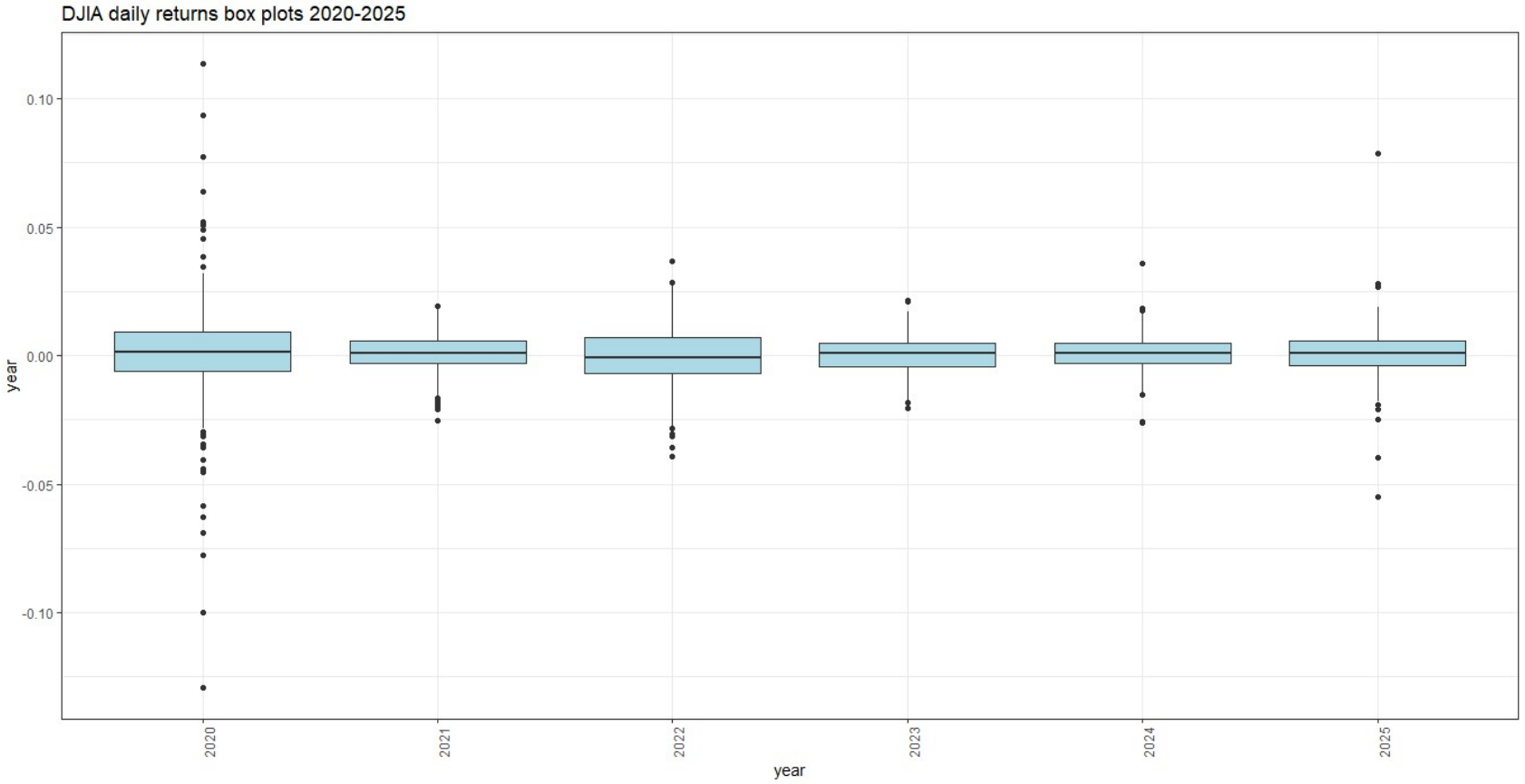


DJI_Adj_Getiri TANIMLAYICI İSTATİSTİKLER

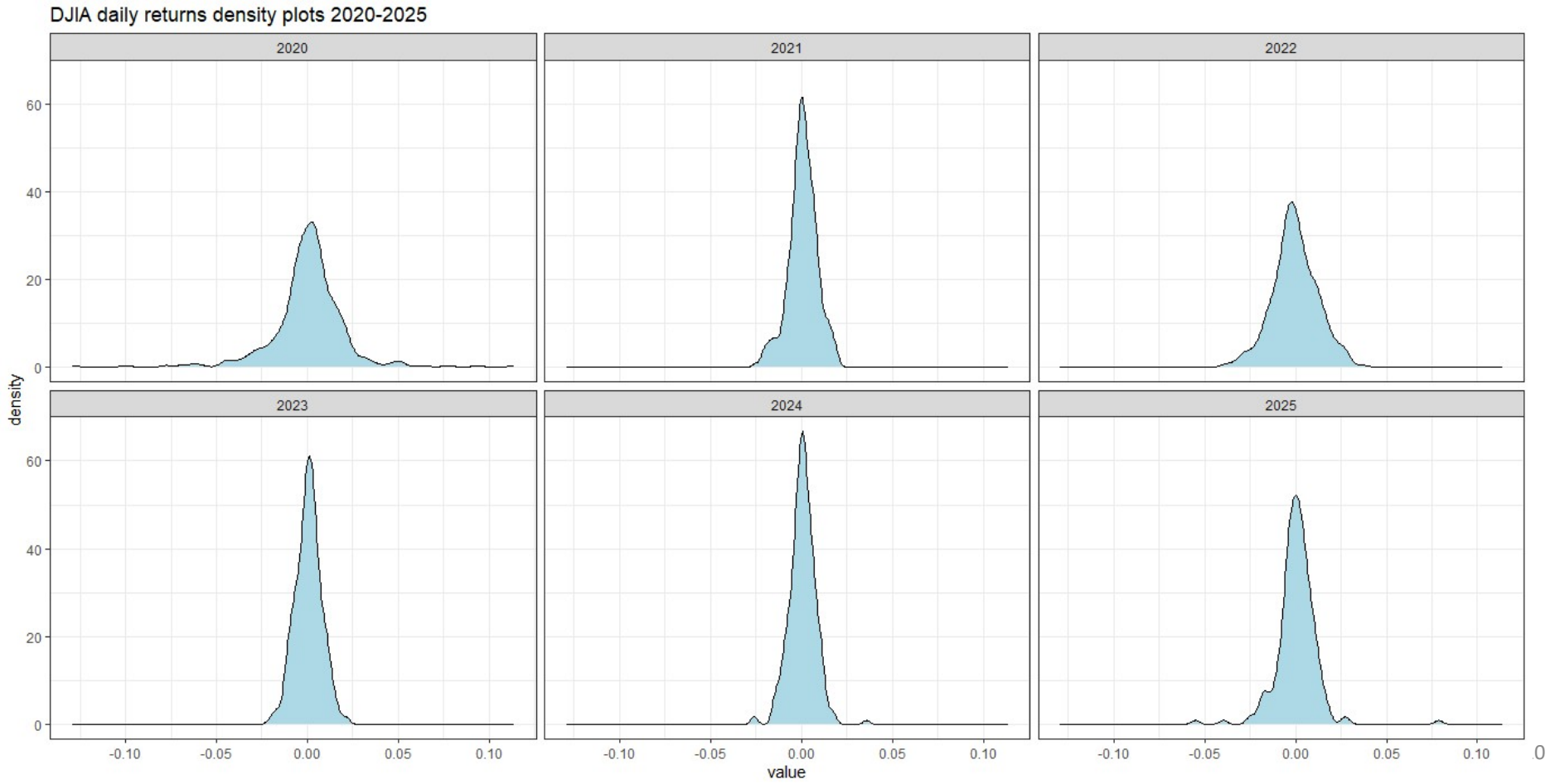
- Yıl bazında DJI_adj_getiri değişkenine ait tanımlayıcı istatistikler

```
> (dj_stats <- dataframe_basicstats(dj_ret_df))
                2020        2021        2022        2023        2024        2025
nobs       252.000000  252.000000  251.000000  250.000000  252.000000  200.000000
NAs          0.000000    0.000000    0.000000    0.000000    0.000000    0.000000
Minimum     -0.129265   -0.025277   -0.039417   -0.020608   -0.026021   -0.055026
Maximum      0.113650    0.019499    0.036951    0.021273    0.035717    0.078704
1. Quartile -0.006233   -0.003097   -0.006911   -0.004160   -0.003174   -0.003846
3. Quartile  0.009067    0.005636    0.007146    0.004738    0.004747    0.005627
Mean         0.000502    0.000712   -0.000288    0.000540    0.000506    0.000529
Median       0.001470    0.000727   -0.000969    0.000746    0.000819    0.000764
Sum          0.126604    0.179423   -0.072385    0.134880    0.127626    0.105821
SE Mean      0.001461    0.000493    0.000788    0.000454    0.000451    0.000793
LCL Mean    -0.002375   -0.000259   -0.001841   -0.000355   -0.000382   -0.001035
UCL Mean     0.003380    0.001683    0.001264    0.001435    0.001395    0.002094
Variance     0.000538    0.000061    0.000156    0.000052    0.000051    0.000126
Stdev        0.023191    0.007830    0.012487    0.007185    0.007163    0.011221
Skewness    -0.458852   -0.353651   -0.051030   -0.011804    0.051460    0.793564
Kurtosis     8.350874    0.653591    0.374704    0.139789    2.701244   13.890744
```

Yıllara göre kutu grafiği



Yıllara göre Olasılık Yoğunluk Grafikleri



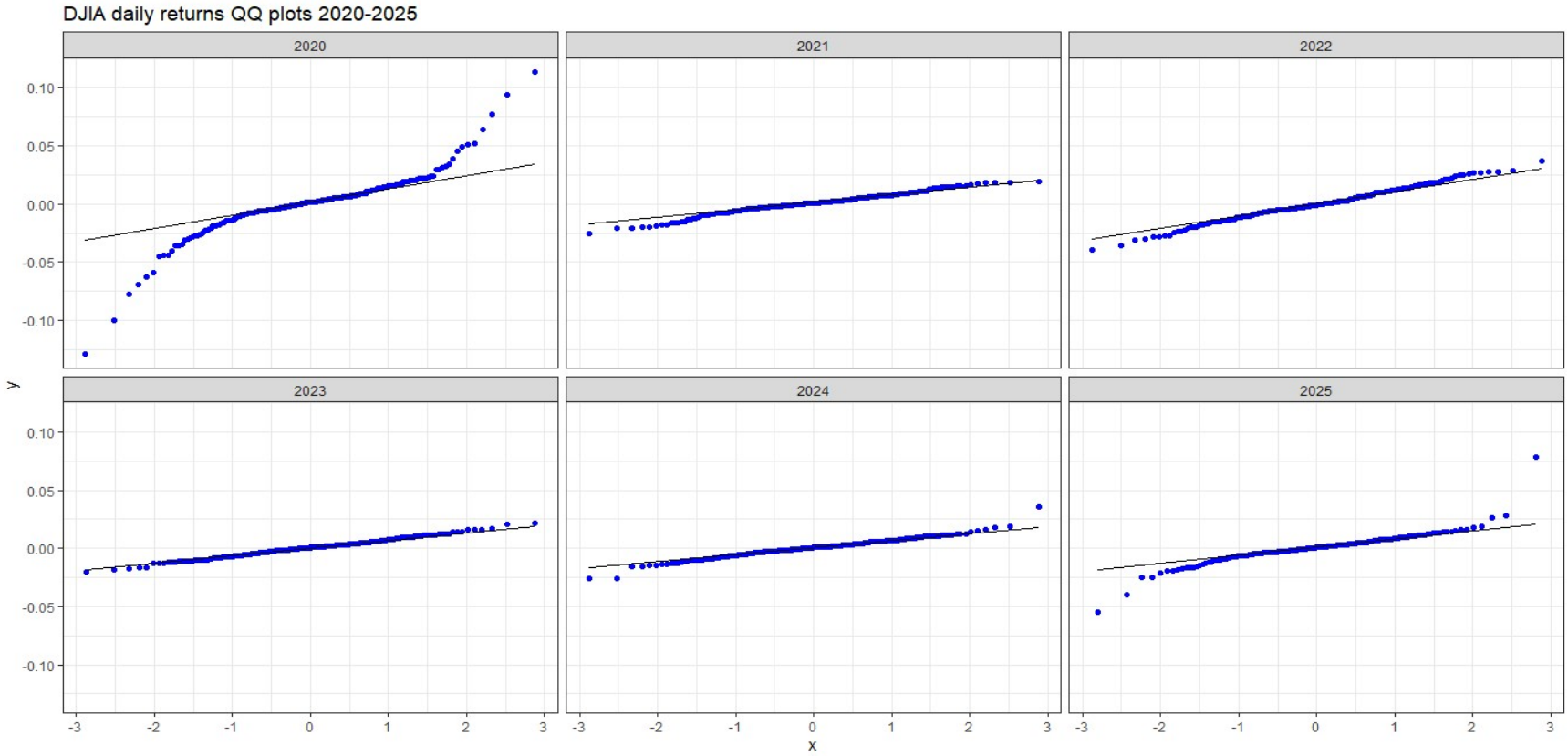
Shapiro-Wilk Normallik Testi

- H0: Veriler normal dağılan bir kitleden gelmektedir.

```
> dataframe_shapirotest(dj_ret_df)
           result
2020 5.746507e-15
2021 2.167066e-03
2022 2.871287e-01
2023 7.812436e-01
2024 4.569052e-05
2025 2.551237e-13
```

- Yıllara göre veriler normal dağılan bir kitleden gelmemektedir.

DJI_Adj_Getiri yıllara göre Q-Q Plot



Ortalamalar Arası Fark Testi

H0: Veriler normal dağılan bir kitleden gelmektedir

```
> shapiro.test(data)

        Shapiro-Wilk normality test

data:  data
W = 0.82685, p-value < 2.2e-16
```

H0 red, normal kitleden gelmemektedir.

- T-test
- H0: Ortalamalar arası fark sıfırdır.
- Normal dağılıma uysaydı,

```
> t.test(sample1,sample2)

        Welch Two Sample t-test

data:  sample1 and sample2
t = -0.21192, df = 1112.1, p-value = 0.8322
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.001464464  0.001178952
sample estimates:
   mean of x    mean of y
0.0003417299 0.0004844855
```

H0 reddedilemez, ortalamalar arası fark istatistiksel olarak önemli değildir, sıfırdır.

Varyans Homojenliği Testi

- H0: İki grubun varyansları eşittir.

```
> var.test(sample1,sample2)

        F test to compare two variances

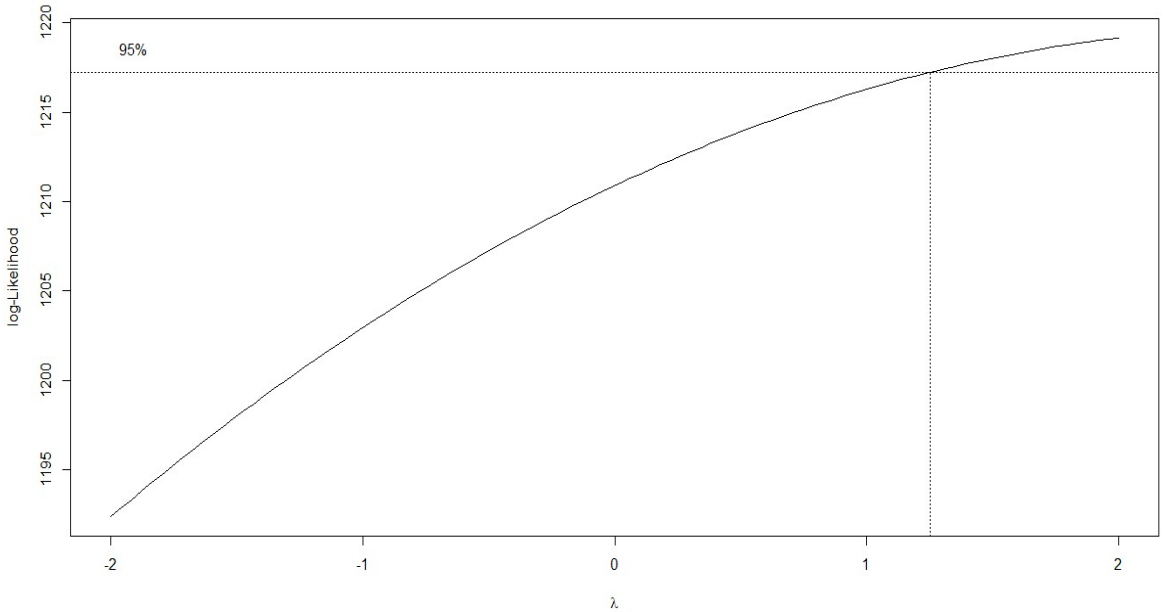
data:  sample1 and sample2
F = 3.4859, num df = 727, denom df = 728, p-value < 2.2e-16
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 3.014009 4.031600
sample estimates:
ratio of variances
          3.485856
```

H0 red, iki grubun varyansı eşit değildir.

Varyans Homojenliği için Box-Cox dönüşümü

```
> data_pos <- data + abs(min(data)) + 1
> b <- boxcox(lm(data_pos ~ 1))
> lambda <- b$x[which.max(b$y)]
> lambda
[1] 2
```

- Orijinal verinin karesi
- Alınırsa varyans homojen
- -liği sağlanabilir.



Trend varlığı için test –cox-stuart test

```
> trend.test(data, method = c("cox.stuart", "diff.sign"), plot = TRUE)

        Approximate Cox-Stuart trend test

data:  data
D+ = 368, p-value = 0.3834
alternative hypothesis: data have a increasing trend
```

- H0: Veride artan trend yoktur.
- H0 REDDEDİLEMEZ. Veride artan trend yoktur.

Trend varlığı-Mann Kendall test

- H0: Trend yoktur.

```
> MannKendall(data)    ###Trend
tau = -0.00726, 2-sided pvalue =0.67792
```

- H0 reddedilemez. Trend yoktur.

Durağanlık Testi- Birim Kök Testi

- Augmented Dickey Fuller Testi
- H0: Seri durağan değildir.

```
Type 1: no drift no trend
     lag   ADF p.value
[1,]   0 -45.4    0.01
[2,]   1 -25.6    0.01
[3,]   2 -21.4    0.01
[4,]   3 -20.9    0.01
[5,]   4 -17.1    0.01
[6,]   5 -17.7    0.01
[7,]   6 -14.0    0.01
[8,]   7 -14.2    0.01
```

- Seride ortalama ve trend yoksa, Type 1 sonuçları yorumlanır.
- Bu sonuçlara göre tüm lag (Gecikmeler) için H0 reddedilir. Yani seri durağan değildir.

Durağanlık Testi- Birim Kök Testi

- Augmented Dickey Fuller Testi
- H0: Seri durağan değildir.

```
Type 2: with drift no trend
     lag   ADF p.value
[1,]   0 -45.4    0.01
[2,]   1 -25.7    0.01
[3,]   2 -21.4    0.01
[4,]   3 -21.0    0.01
[5,]   4 -17.1    0.01
[6,]   5 -17.8    0.01
[7,]   6 -14.1    0.01
[8,]   7 -14.3    0.01
```

- Seride ortalama var, ama trend yoksa, Type 2 sonuçları yorumlanır.
- Bu sonuçlara göre tüm lag (Gecikmeler) için H0 reddedilir. Yani seri durağan değildir.

Durağanlık Testi- Birim Kök Testi

- Augmented Dickey Fuller Testi
- H0: Seri durağan değildir.

```
Type 3: with drift and trend
     lag   ADF p.value
[1,]   0 -45.4    0.01
[2,]   1 -25.6    0.01
[3,]   2 -21.4    0.01
[4,]   3 -21.0    0.01
[5,]   4 -17.1    0.01
[6,]   5 -17.8    0.01
[7,]   6 -14.1    0.01
[8,]   7 -14.3    0.01
```

- Seride hem ortalama, hem trend varsa, Type 3 sonuçları yorumlanır.
- Bu sonuçlara göre tüm lag (Gecikmeler) için H0 reddedilir. Yani seri durağan değildir.